When Labels Override Evidence

Causal pathways for label shortcuts, option interaction and confidence in a non-autoregressive decision model

2026-10-09 · Vignesh Kanike · model: convaiinnovations/laya (laya 0.4.1)

Laya scores each answer option at its own [MASK] token in one encoder pass. Its yes/no answers follow the *meaning* of the option label words instead of the passage: swapping "yes"/"no" flips 96% of correct answers, while swapping "A"/"B" flips 3%. We trace this to a direct, redundant read of the label by the [MASK] in local attention layers, separate from an evidence route that runs through the question tokens. A one-edge guard removes the shortcut on untouched human-written data, and controlled retraining shows which kind of training data creates it. Order sensitivity and confidence are handled by separate, simpler mechanisms.

At a glance

Claim	Verdict	Key number
C3 · label shortcut (headline)	Confirmed	yes/no swap: 96% follow the label; A/B: 3%
C3 · mechanism	Two pathways	direct [MASK]→label read, redundant across local layers; evidence via question tokens (restoring them recovers 0.91)
C3 · confirmatory benchmark	Confirmed (frozen, human-written)	yes/no label-following 96.7% (n = 516); guard 2.6%; same-size control edges 97.1%
C3 · guard	Confirmed on untouched data	RTE 96% → 4%; MultiRC 92% → 6%
C3 · origin	Shown by retraining (6 vs 6 seeds)	planted 88.9% vs baseline 14.7% on the frozen benchmark; exact permutation p = 0.0022; a weaker but sole-cue label (90%) is enough
C1 · order vs competition	Separable	blocking option interaction removes 8.0 [4.1, 12.0] points of flips; order-invariance keeps 78% of competition
C2 · confidence	Logit amplifier, not regulator	late-MLP ablation ECE 0.080; per-item rescale 0.082; best temperature 0.030
C4 · jailbreak	Supporting	AUROC 0.947; decodable at layer 0, decided late
C5 · CE vs RLCD	Trend only	competition 0.446 vs 0.374, p = 0.09 (3 seeds each)
Replication	3 checkpoints	laya, laya-typed-decisions, laya-multilingual (different base)

---

1 · Model and instrument

Laya is ModernBERT-large (28 layers, width 1024, 16 heads; global attention on every third layer from layer 0, a ±64-token window elsewhere) plus two transformer head layers and a per-option scorer. The input is `[CLS] <type> question: <instructions> [SEP] [MASK] option0 [MASK] option1 … [SEP] state [SEP]`. A noul (yes/no) option renders as "`false: no, the statement does not hold`" / "`true: yes, the statement holds`". All analysis uses an explicit re-implementation of the forward pass with named, editable sites, checked against laya's own `predict`.

Checkpoint	Max probability difference vs predict	Local window	Throughput (H100, bf16, 512 tokens)
laya	5.1e-05	±64	494 seq/s

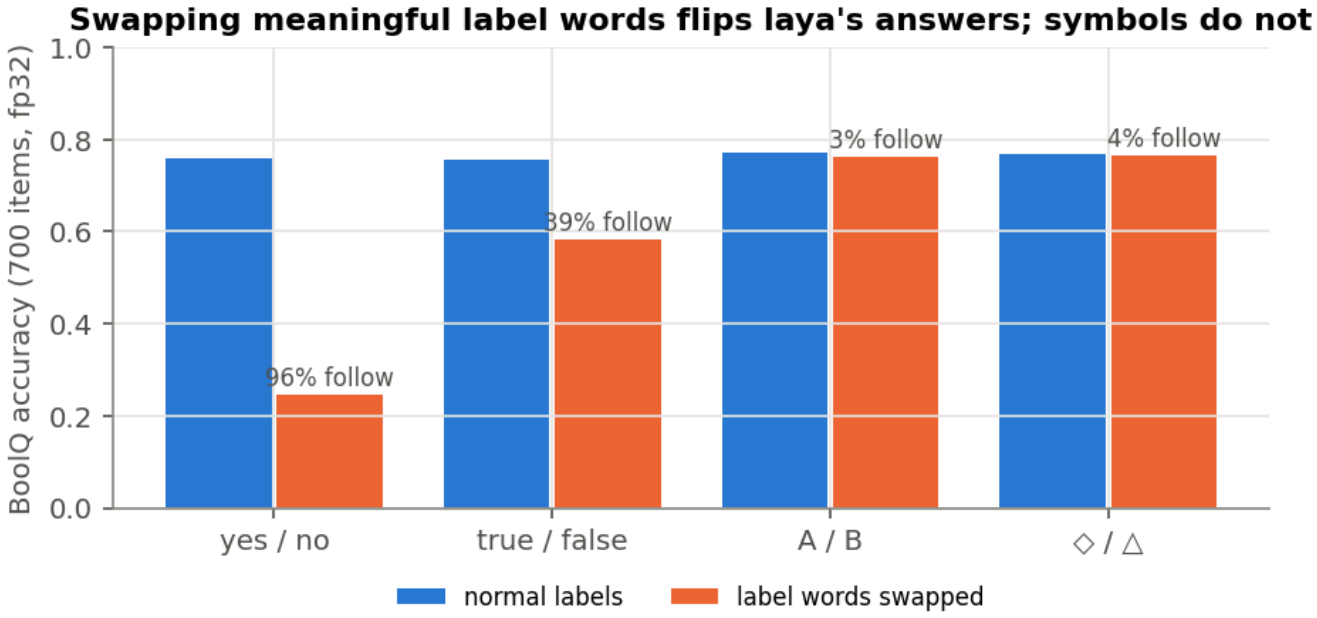
Checkpoint	Max probability difference vs predict	Local window	Throughput (H100, bf16, 512 tokens)
smolaya	5.0e-05	±64	679 seq/s
typed	5.0e-05	±64	343 seq/s
multilingual	4.9e-05	±64	—

Definitions

- **Paired transition.** Each item is scored with normal and with swapped label words; we report counts of correct→correct, correct→wrong, wrong→correct and wrong→wrong.
- **Label-following** = n(correct→wrong) / n(correct with normal labels); the denominator n is reported. Item-level 95% intervals are bootstrap intervals over those n items.
- **Swap gap** = accuracy with normal labels − accuracy with swapped labels, over all items.
- **Recovery** (patching / restoration) per item = (y_patched − y_corrupt) / (y_clean − y_corrupt), y = log-odds of the "holds" option; items with |y_clean − y_corrupt| < 0.5 are excluded and counted; we report the median with a bootstrap interval.
- **Seed-level tests** use the trained model as the unit: per-seed values are listed and compared with an exact two-sided permutation test (and Welch's t-test).
- **Confirmatory vs exploratory.** The frozen benchmark (600 MNLI + QNLI items, SHA-256 cea5547f…) and its interventions were fixed before evaluation; analyses added afterwards are labelled exploratory.

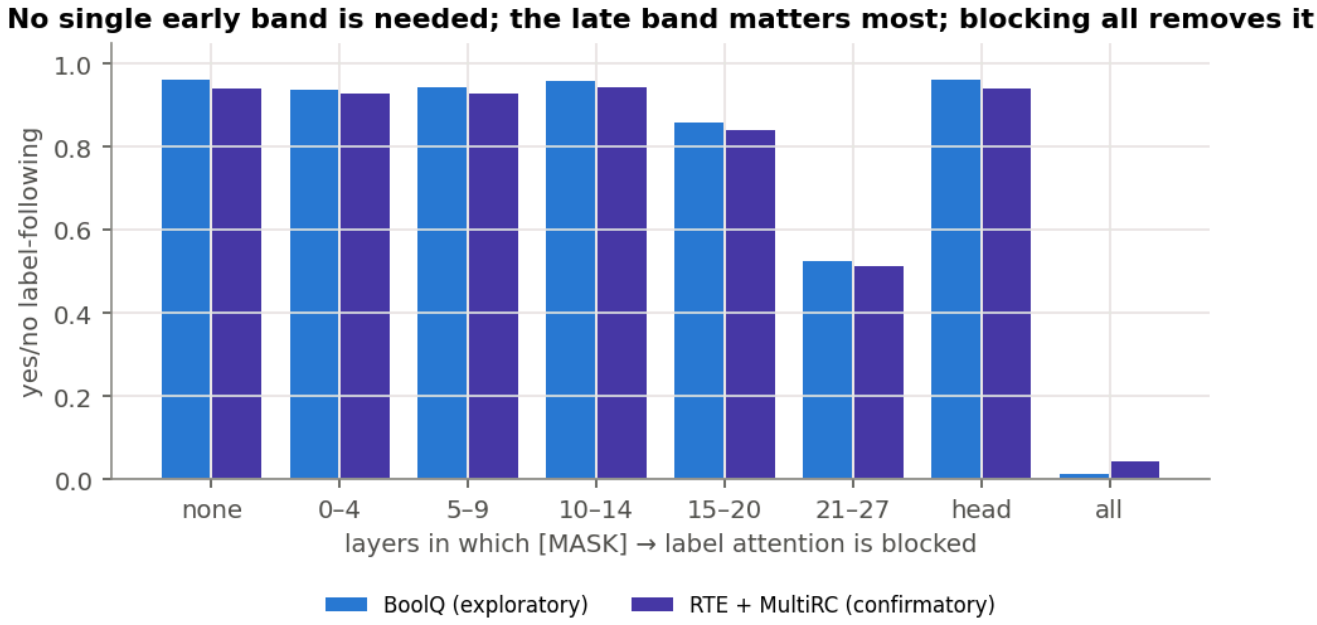
2 · C3: labels override evidence

2.1 Behaviour

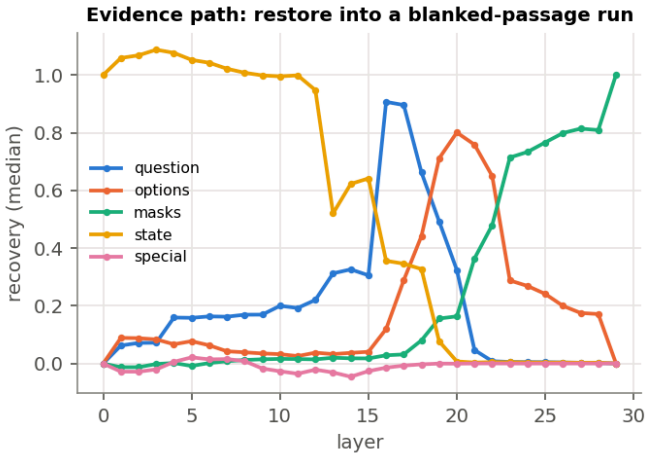


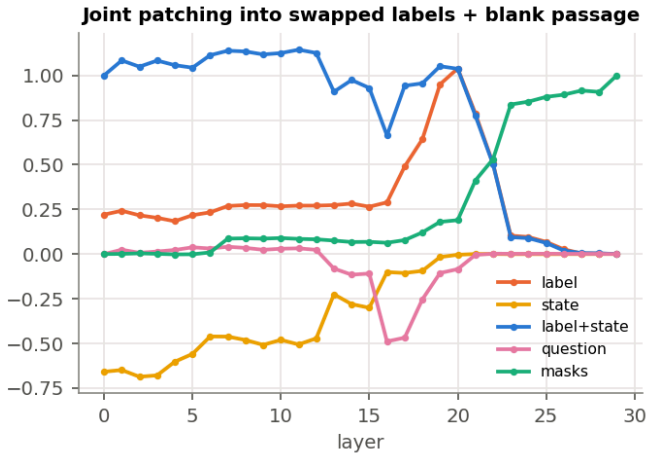
Test: each option's label word is swapped ("true: no, the statement does not hold"). A model reading the passage is unaffected; a model reading the label word picks the option the passage contradicts. Label-following is the share of items answered correctly with normal labels that flip when the words are swapped. The effect replicates on 300 held-out BoolQ items and on GPT-written items, and laya's model card documents the failure; our contribution is the mechanism and fix.

2.2 Two pathways



Block reading of the passage by...	Accuracy, plain	Accuracy, with guard
none	0.737	0.727
masks	0.730	0.733
options	0.760	0.757
question	0.537	0.507
special	0.747	0.747
all_nonstate	0.517	0.447



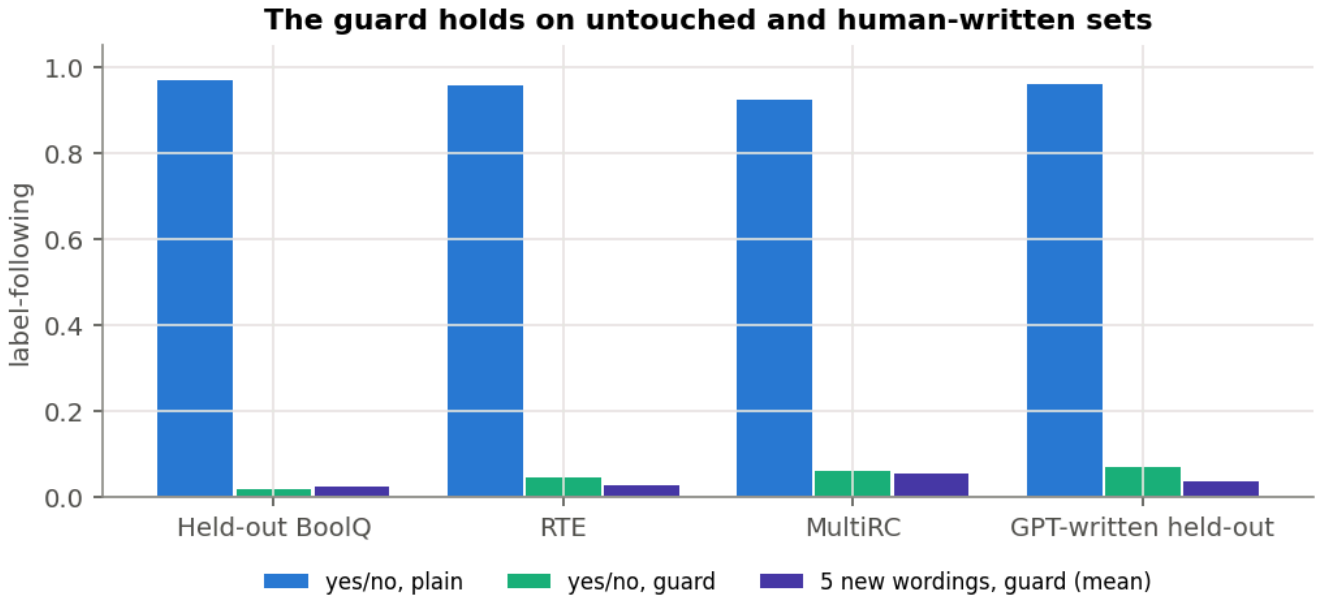


- **Label read is redundant and mostly late.** Blocking [MASK]→label in layers 10–14 alone leaves label-following at 96%; blocking layers 21–27 alone halves it (53%); blocking it everywhere except any single band from 0–20 removes it. An earlier account placing the read at layers 10–14 was wrong and is withdrawn.
- **Evidence goes through the question tokens.** Stopping them from reading the passage drops accuracy from 0.737 to 0.537; stopping the [MASK]s or option text from reading it changes nothing. Restoring only the question tokens at layer 16 recovers 0.91 of the decision.
- **No timing race.** Label information leaves the label tokens around layer 20 and enters the [MASK] at layers 21–23, the same layers where the evidence arrives.
- **Contradicting passages** (187 BoolQ passages edited by gpt-5.6-terra so the answer flips, checked by gpt-5.6-luna): with consistent labels laya follows the edited passage 71% of the time, with swapped labels 32%. With the passage blanked,

swapping labels moves "holds" answers from 22% to 91%.

2.3 The guard

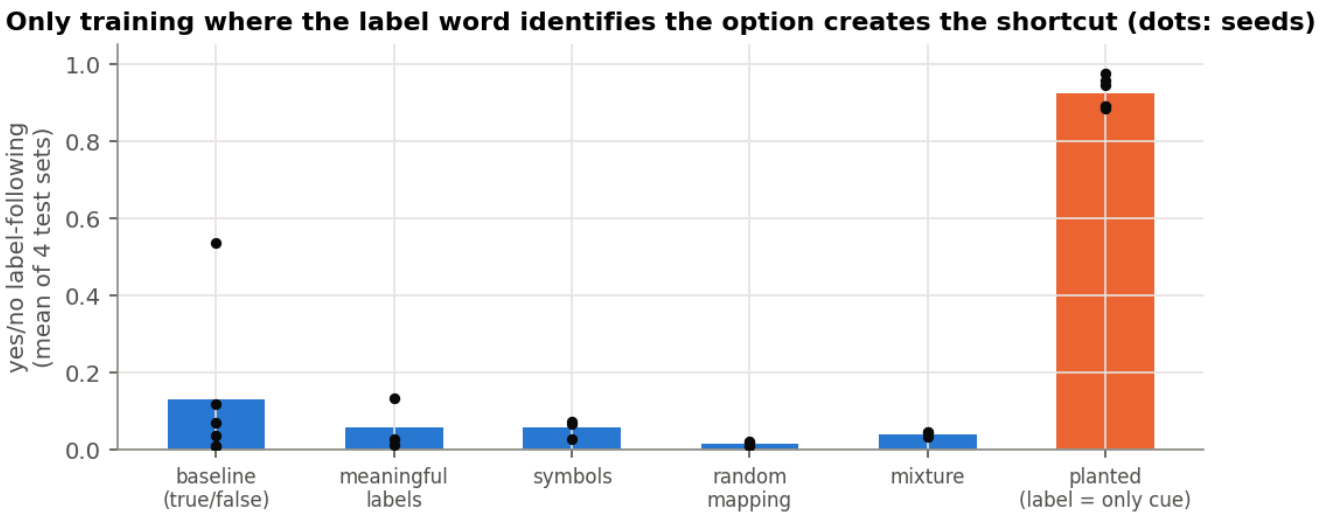
The guard blocks noul [MASK]s from attending to their label words in local layers only, installed on a stock `laya.load` agent and run through the official `predict` API (`interp/label_guard.py`). On BoolQ yes/no it cuts label-following from 95.9% to 1.3%. Choice questions are bit-identical (max difference 0); jailbreak AUROC moves 0.940 → 0.919 (1.8% of decisions change); injection AUROC 0.898 → 0.870.



Untouched set (fp32)	yes/no following, plain → guard	Accuracy, plain → guard	5 new wordings (mean)	Multi-token labels (extended guard)
Held-out BoolQ (300)	97% → 2%	0.757 → 0.753	74% → 2%	96% → 32% (4%)
RTE (277, human-written)	96% → 4%	0.780 → 0.812	80% → 3%	97% → 33% (13%)
MultiRC (500, human-written)	92% → 6%	0.586 → 0.614	80% → 5%	93% → 37% (13%)
GPT-written held-out (292)	96% → 7%	0.767 → 0.808	73% → 4%	96% → 37% (5%)

Multi-token labels ("yes, it does") are the guard's failure mode. The bracketed extended guard also blocks option text from reading the labels; it was designed after seeing these sets and is exploratory.

2.4 Where the shortcut comes from



Training regime (RLCD, 3 seeds)	yes/no label-following	With guard	Accuracy
Baseline: default true/false	12.8% ± 20.3	2.6%	0.749
Meaningful labels, aligned	5.8% ± 6.6	1.3%	0.751
Meaningless symbols	5.5% ± 2.4	1.2%	0.754
Meaningful labels, randomly swapped	1.4% ± 0.6	1.0%	0.752
Mixture	4.0% ± 0.6	2.2%	0.756
Planted: label word is the only cue	92.3% ± 4.0	47.2%	0.768

Fifteen models were trained from ModernBERT-large on the same 31,419 items with laya's loss; only the yes/no labels differed. Only training where the label word is the sole cue to which option means what (shuffled order, identical option text) creates the shortcut ($p = 0.000$ vs baseline, seeds as units); meaningful labels alone do not ($p = 0.46$). The guard's effect on the planted models differs by seed (28%, 80%, 13%, 59%, 27%, 77%): the same behaviour is implemented by different routes, so a fix derived from one model's internals can fail on a retrained sibling. Laya's own training data are not public, so this shows what can create the shortcut, not what did.

2.5 Replication

Checkpoint	RTE yes/no, plain → guard	RTE multi-token	Following after blocking only L21–27	Accuracy when question tokens cannot read passage	Best question-token restoration
laya	96% → 4%	97% → 33%	51%	0.737 → 0.537	0.91
typed-decisions	93% → 3%	96% → 14%	23%	0.760 → 0.620	0.91
multilingual	91% → 1%	95% → 5%	90%	0.670 → 0.587	0.75

The guard and the two-pathway picture hold on all three checkpoints; where the label is read in depth, and how clean the question-token relay is, vary by model.

2.6 Confirmatory benchmark and matched controls

A new benchmark of 600 human-written items (300 MNLI, 300 QNLI; never used before) was frozen and hashed before any evaluation, with 519 contradicting edits (gpt-5.6-terra, checked by gpt-5.6-luna) and three never-used wordings plus two multi-token ones. On laya, yes/no label-following is 96.7% ($n = 516$, 95% CI 95.2%–98.1%); transitions: correct→correct 17, correct→wrong 499, wrong→correct 70, wrong→wrong 14.

Intervention (laya, fp32)	yes/no	right/wrong (new)	"yes, that is right" (new)	◇/△	Accuracy	Jailbreak / injection AUROC
None	96.7%	80.4%	96.9%	4.1%	0.860	0.940 / 0.896
Guard: [MASK]→label, local layers	2.6%	3.0%	28.8%	2.0%	0.842	0.917 / 0.872
Extended guard: + option text→label	1.8%	1.6%	7.6%	2.8%	0.852	0.911 / 0.875
Control: same number of random option edges	97.1%	85.7%	98.2%	4.5%	0.858	0.940 / 0.897
Control: same number of description edges	97.9%	85.1%	98.2%	5.4%	0.855	0.939 / 0.895
Control: [MASK]→[CLS]	98.3%	91.9%	99.2%	4.1%	0.867	0.946 / 0.897
Evidence path blocked: question→passage	84.2%	72.6%	81.6%	10.6%	0.645	0.881 / 0.742

Checkpoint	yes/no, none	Guard	Extended guard, multi-token	Random-edge control	Accuracy, none → guard
laya	96.7%	2.6%	7.6%	97.1%	0.860 → 0.842
typed	93.5%	3.9%	2.8%	95.7%	0.825 → 0.845
multilingual	88.7%	1.6%	6.5%	89.0%	0.735 → 0.743

- The guard is specific: blocking the same number of other edges, or a sham edge, leaves label-following unchanged.
- The extended guard, designed earlier on other sets, passes this first confirmatory test, including multi-token labels.
- On contradicting passages the guard cuts label-following from 97.5% to 4.5%. With the passage blanked, swapping labels moves "holds" answers from 33% to 95%; with the guard, 5% to 11%.
- Recovery from restoring the question tokens peaks at layer 16: median 0.91 [0.85, 0.96], n = 275 (25 of 300 excluded for a small gap).

2.7 Dose response, guard scope and an external architecture

Training arm (RLCD; shuffled order except baseline)	yes/no label-following per seed (%)	Mean (%)	With local guard per seed (%)	Accuracy
Baseline: default labels, fixed order	0.2, 0.7, 18.1, 9.3, 54.9, 4.7	14.7	2.2, 0.2, 2.7, 3.1, 3.7, 3.3	0.773
D0: label random, informative text	1.3, 0.0, 0.9	0.8	2.2, 0.4, 1.1	0.743
D1: label always right, informative text	2.8, 31.2, 98.5	44.2	3.0, 7.3, 92.7	0.775
D2: label right 75%, neutral text	91.9, 41.1, 90.9	74.6	44.2, 58.7, 59.8	0.688
D3: label right 90%, neutral text	95.3, 92.0, 85.8	91.0	57.5, 79.7, 32.2	0.764
Planted: label always right, neutral text	81.2, 86.3, 84.2, 94.7, 92.3, 95.0	88.9	33.3, 78.1, 16.1, 65.5, 28.8, 71.1	0.784

Planted vs baseline with 6 seeds each: exact two-sided permutation p = 0.0022 (the smallest value this test allows: every planted model exceeds every baseline model). A sole-cue label that is right only 75% or 90% of the time already produces strong label-following. When the label is redundant with informative option text (D1), reliance on it depends on the seed (one seed of three at 98%). Baseline models are not immune on this benchmark (one of six seeds at 55%).

Model (exploratory, post hoc)	None	Guard, local layers	Guard, all layers	Extended guard, all layers
laya	96.7% (0.860)	2.6% (0.842)	2.6% (0.838)	0.6% (0.843)
p5_planted_s0	81.2% (0.778)	33.3% (0.777)	2.0% (0.738)	0.7% (0.762)
p5_planted_s1	86.3% (0.817)	78.1% (0.777)	26.8% (0.610)	19.1% (0.602)

Model (exploratory, post hoc)	None	Guard, local layers	Guard, all layers	Extended guard, all layers
p5_planted_s2	84.2% (0.780)	16.1% (0.765)	5.3% (0.758)	1.8% (0.753)
p5_planted_s3	94.7% (0.783)	65.5% (0.758)	7.1% (0.730)	4.6% (0.725)
p5_planted_s4	92.3% (0.780)	28.8% (0.777)	3.6% (0.733)	2.2% (0.748)
p5_planted_s5	95.0% (0.763)	71.1% (0.727)	12.7% (0.552)	0.3% (0.623)

Label-following (accuracy). The local-only guard's uneven effect on retrained models is mostly a scope issue: blocking the same [MASK]→label edges in every layer removes most label-following in most seeds, so the retrained models use the same edge class but in different layers. In two seeds the all-layer block also costs 20+ accuracy points: those models rely on the label even to answer normally.

External model (RoBERTa-large, option-marker head)	BoolQ accuracy	Label-following	With [MASK]→label block	Accuracy, none → question→passage blocked
RoBERTa planted_s0	0.837	97.4%	1.5%	0.640 → 0.568
RoBERTa planted_s1	0.643 (did not learn)	99.7%	5.8%	0.540 → 0.540
RoBERTa planted_s2	0.770	84.9%	1.2%	0.597 → 0.557
RoBERTa baseline_s0	0.633 (did not learn)	0.3%	0.0%	0.543 → 0.543
RoBERTa baseline_s1	0.837	42.1%	4.2%	0.772 → 0.597
RoBERTa baseline_s2	0.627 (did not learn)	10.8%	0.0%	0.542 → 0.543

RoBERTa replicates the training-origin result and the block's effect in the runs that learned the task, but 3 of 6 runs did not learn it (laya's learning rate is tuned for ModernBERT), so this is a partial replication.

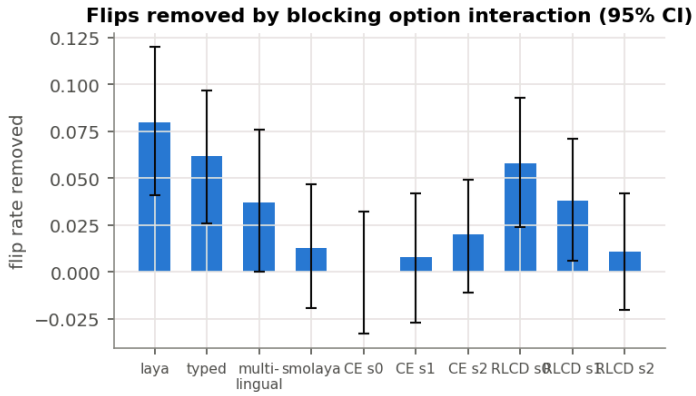
---

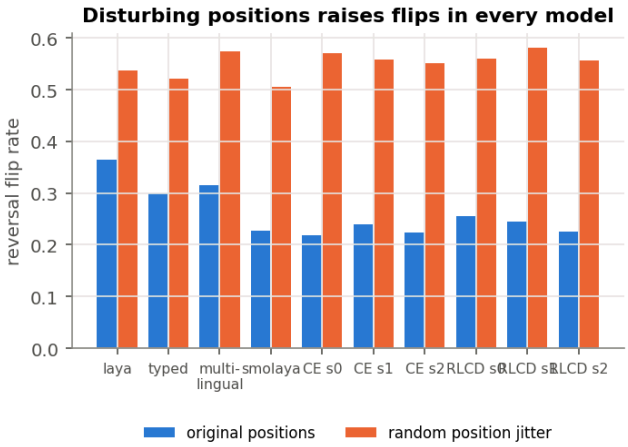
3 · C1: order sensitivity is positional; competition is separate

Reversing the options flips 29.6% of laya's multiple-choice answers (fp32, 1,392 items). Options do compete: blanking one shifts another's centred logit by 0.37 on average. Slot index explains 0.2% of order-induced variance, so this is not a simple primacy bias.

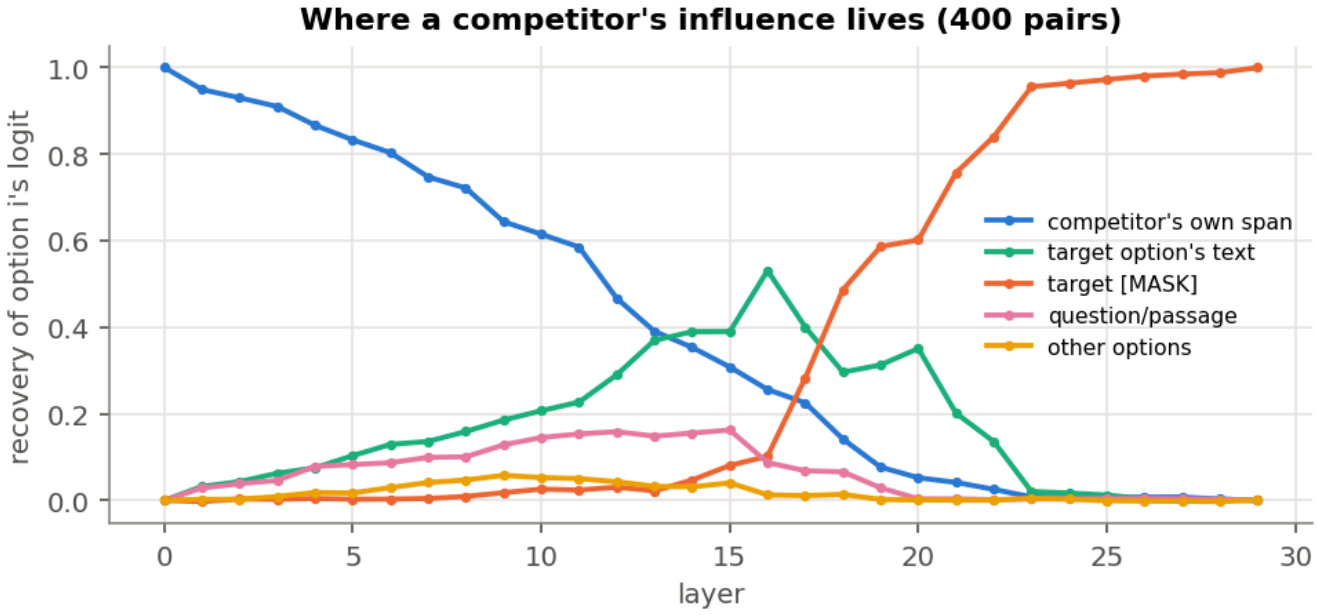
Condition (laya, fp32)	Accuracy	Reversal flips	Mean abs(M)
Trained layout	0.453	29.6%	0.374
Options cannot see each other	0.414	28.3%	0.181
Options blocked + shared positions	0.408	0.0%	0.175
Fully masked parallel	0.346	28.6%	0.000

Position binds each [MASK] to its option. Sharing positions while leaving attention unrestricted makes every [MASK] identical (same token, position and keys), so all options score the same (accuracy 0.246). The factorial therefore holds binding fixed (each [MASK] reads only its own option) and varies option-text interaction × positions. With shared positions and symmetric masks the model is order-invariant by construction; the measured quantities are the accuracy cost and how much competition survives.





Model (confirmatory set)	Flips removed by blocking interaction (pts, 95% CI)	Competition kept when order-invariant	Accuracy cost (pts)	Flips under random jitter
laya	8.0 [4.1, 12.0]	78%	1.9	54%
typed	6.2 [2.6, 9.7]	76%	1.4	52%
multilingual	3.7 [0.0, 7.6]	76%	-0.7	57%
smolaya	1.3 [-1.9, 4.7]	77%	4.2	50%
CE s0	0.0 [-3.3, 3.2]	86%	9.7	57%
CE s1	0.8 [-2.7, 4.2]	87%	9.6	56%
CE s2	2.0 [-1.1, 4.9]	90%	8.9	55%
RLCD s0	5.8 [2.4, 9.3]	77%	8.2	56%
RLCD s1	3.8 [0.6, 7.1]	82%	8.6	58%
RLCD s2	1.1 [-2.0, 4.2]	87%	10.4	56%



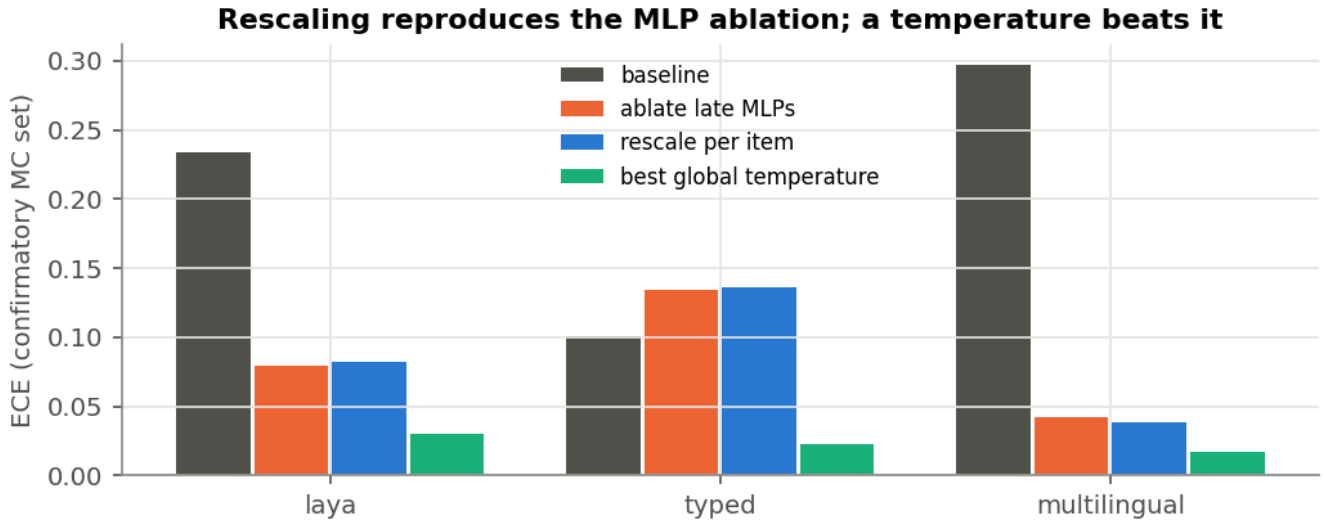
Position permutation (Phase 8): giving the options reversed position ids without moving the tokens reproduces the reversed run in 100% of flips in all 10 models. This holds by construction, since a bidirectional encoder sees only (token, position) pairs, so order effects are positional by definition; the substantive result is the factorial one above. The competition measure is robust to the replacement used (laya: filler 0.39, dash 0.40, distractor 0.44).

A competitor's content builds up in the target option's text tokens through about layer 16, then enters the target [MASK] at layers 17–22; no small set of heads carries it.

4 · C2: a late-MLP logit amplifier

Removing MLPs 22, 27, 26 halves laya's logit scale (ratio 0.47) and fixes most miscalibration, but per-item rescaling reproduces it and a single temperature beats it, so by the pre-registered rule these MLPs are reported as an amplifier, not a confidence regulator. The effect sits entirely at the [MASK] positions. Unlike the entropy neurons described for decoder LLMs, which the authors distinguish from a temperature, these behave like one.

Laya, confirmatory MC set (fp32)	Accuracy	ECE	Brier	NLL	Answers kept
Baseline	0.364	0.234	0.816	1.646	100.0%
Ablate late MLPs	0.372	0.080	0.742	1.393	95.8%
Rescale each item to the ablated size	0.364	0.082	0.739	1.385	100.0%
One temperature, same mean scale	0.364	0.078	0.733	1.361	100.0%
Best global temperature	0.364	0.030	0.719	1.329	100.0%
Ablate only at the [MASK]s	0.372	0.078	0.741	1.391	95.7%
Ablate everywhere except the [MASK]s	0.365	0.235	0.816	1.646	99.1%



The scale ratio barely depends on the input (coefficient of variation 0.18; correlation with correctness +0.08; with the original margin +0.44). In typed-decisions and multilingual the selected late MLPs carry nearly the whole logit scale (ratios 0.003 and 0.007); the same rescaling verdict holds. With the temperature fitted on the exploratory split ($T = 2.61$) and applied to the confirmatory split, ECE is 0.055 vs 0.080 for MLP ablation.

5 · C4: jailbreak and injection (supporting)

- Detection AUROC 0.947 on natural jailbreaks, 0.867 on injections.
- A linear probe reads the attack from layer 0 at 0.92; patching shows the decision carried by the prompt until layer 15, relayed by question tokens, and completed in the head layers: the early-decodable, late-decisive pattern reported for decoder LLMs (Wen et al. 2026), here in an encoder.
- Adding the jailbreak direction at the layer-20 [MASK]s raises benign prompts flagged from 23% to 61% with an unrelated question at 3%.
- After gpt-5.6-terra removed only the override from 145 flagged jailbreaks, laya still flags 82%: it reacts to jailbreak style more than to the instruction (edits not hand-checked).
- Style test: the same 150 harmless requests are flagged 12.7% written plainly and 54.0% written in jailbreak style (persona/role-play, no override); for the injection question 2.7% vs 42.7%. The pairs are GPT-written and await a human audit.
- Hindi (laya-multilingual): projecting out the English-fit direction drops detection of translated jailbreaks from 52% to 8.5%.

6 · C5: training objective

3 seeds per arm	Cross-entropy	RLCD (laya's loss)	Welch p
Competition, mean abs(M)	0.446 ± 0.041	0.374 ± 0.036	0.09
Calibration error	0.129 ± 0.015	0.103 ± 0.022	0.18
Accuracy	0.709 ± 0.010	0.695 ± 0.007	0.12

RLCD trends toward less competition and better calibration; neither difference is significant with 3 seeds. The position effect, the late-MLP amplifier and the absence of label-following appear under both losses.

7 · Related work

Order and option-ID bias are documented for decoder LLMs (Pezeshkpour & Hruschka 2024; Zheng et al. 2024), and Anchored Answers traces GPT-2's preference for "A" to MLP value vectors. Set-Based Prompting removes order dependence by sharing positions and masking attention; its ablation on 20 MMLU questions found the mask more important than positions in decoders, the opposite of what blocking attention alone does in laya. Label Words are Anchors studies in-context demonstration labels in decoders, a different setting from answer-option labels. Confidence Regulation Neurons describes entropy neurons that are explicitly not a temperature; laya's late MLPs are. Wen et al. (2026, read at abstract level) report early-decodable, late-decisive prompt-injection compliance in decoders. Different seeds learning different circuits is known (A Toy Model of Universality). A BERT cross-encoder BM25 circuit is the closest encoder precedent found. Full notes and the search log are in `LITERATURE_NOTES.md`.

8 · Limitations and corrections

- Laya was not built for multiple-choice QA (its model card notes near-chance zero-shot performance on typed decisions); C1 and C2 describe how it handles options, not tuned QA behaviour.
- Order-invariance is guaranteed by construction in the shared-position cells; positions alone could not be tested.
- The multi-token extended guard is exploratory; contradicting passages and jailbreak rewrites were checked by a second model, not by hand; Wen et al. was read at abstract level only; SAE feature labels were contaminated and are not used.
- Human audits of the GPT-written items, contradicting edits, jailbreak rewrites and style pairs are prepared (`audit/`) but not yet done.
- The all-layer guard results on retrained models are exploratory (added after seeing the local-guard results); the RoBERTa replication is partial (3 of 6 runs did not learn).
- Withdrawn or corrected during the study: the early-timing account of label reading; "a confidence regulator" (now amplifier); a single-seed "20% less competition" (now a non-significant trend); "steering works only late"; "label-following comes from training data" (now: which training can create it); "same behaviour, different mechanisms" (now: same edge class, different layers); the planned "position vs token order" test (an identity by construction, not evidence).

9 · Reproduction and cost

Environment: laya 0.4.1, transformers 5.19.0, torch 2.14.1, nnsight 0.7.0, datasets 5.1.0, Modal client 1.6.1, Python 3.12 on H100 GPUs. Every result file under `release/results` is produced by `modal run modal_interp.py --task <name>` (see `release/README.md`), and this report is rebuilt from those files by `python release/scripts/build_report.py`. Modal compute: 66.16 metered, $0.00 billed at build time. OpenAI usage (gpt-5.6-terra and gpt-5.6-luna) is listed in `release/README.md`.